

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Experiment 5

Student Name: Jashanpreet Kaur
Branch: CSE
Semester: 6th
Subject Name: System Design

UID: 23BCS14043
Section/Group: KRG 2-A
Date of Performance: 26/02/2026
Subject Code: 23CSH-314

1. **Aim:** Design a Messenger Application (Similar to WhatsApp / Facebook Messenger): To design and analyze the architecture of a scalable real-time messenger application that supports one-to-one and group communication with high availability and low latency.

2. Objective:

- To implement real-time message delivery using WebSockets.
- To support one-to-one and group messaging.
- To preserve chat history.
- To ensure high availability and reliability.
- To design APIs for user management and chat operations.
- To design the system for large-scale usage (up to 1 Billion users).

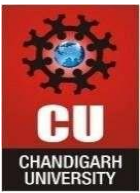
3. Tools Used:

- Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket) o Built responsive UI with real-time communication support
- Backend (Node.js / Java / Spring Boot) o Developed scalable REST APIs & WebSocket server
- Database (MongoDB / MySQL / PostgreSQL) o Managed structured & unstructured data efficiently
- Redis o Implemented caching & pub-sub for high performance
- Deployment (AWS / GCP, Load Balancer, Docker) o Cloud deployment with scalability, containerization & traffic distribution

4. System Requirements:

A. Functional Requirements

- User Registration
 - a. POST /user/register
 - b. Fields: userName, email, phoneNumber
- User Login



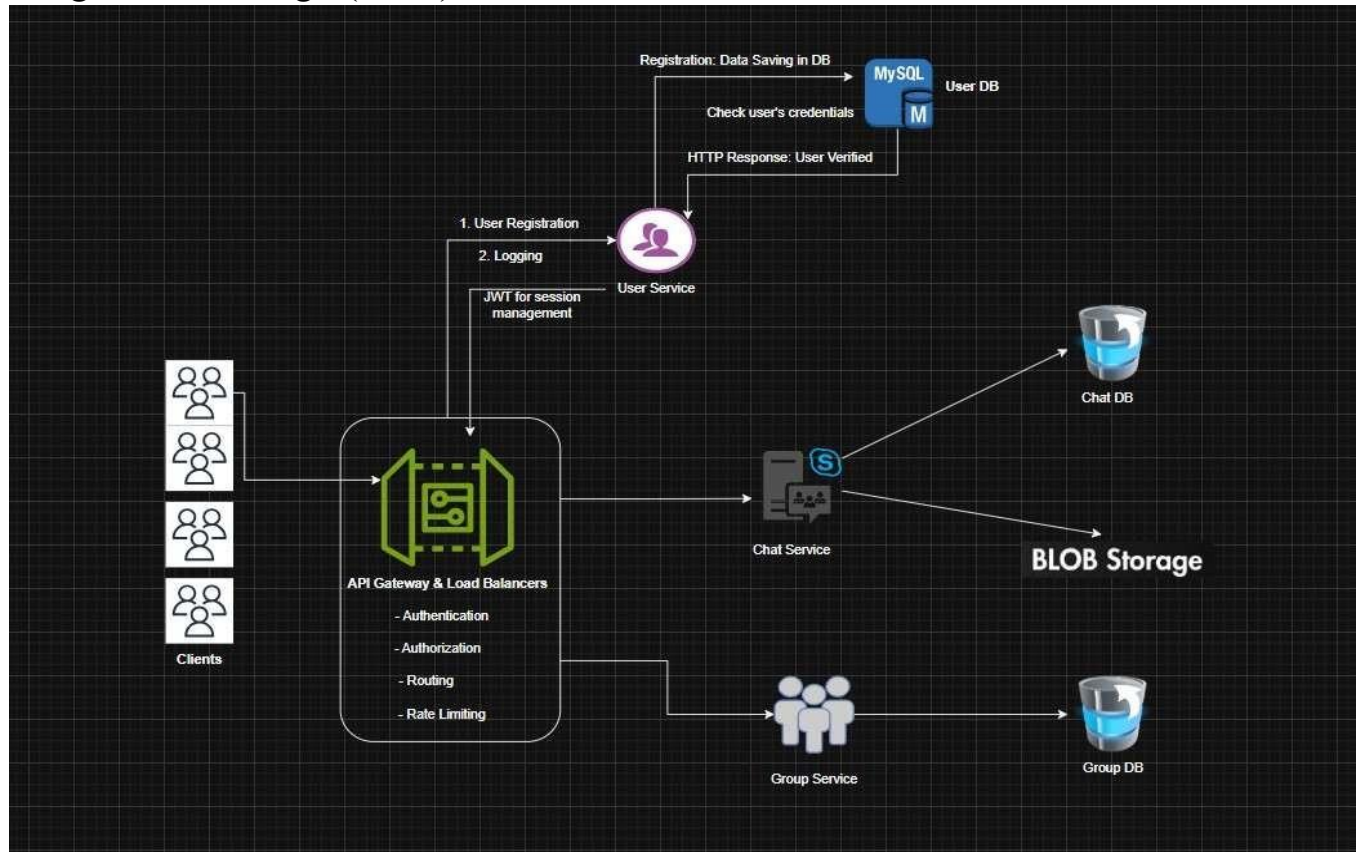
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

- a. POST /user/login
- One-to-One Messaging
 - a. WS: /messages/send
 - b. GET /messages/{userId}/{receiverId}
 - c. Pagination support
- Group Messaging
 - a. POST /groups/create
 - b. POST /groups/{groupId}/add
 - c. DELETE /groups/{groupId}/remove
 - d. WS: /messages/send
- Message Types
 - a. Text messages
 - b. Media messages (images, videos, files)
- Message History
 - a. Persistent storage
 - b. Chat list view
- Read Receipts
 - a. Delivered
 - b. Seen

B. Non-Functional Requirements

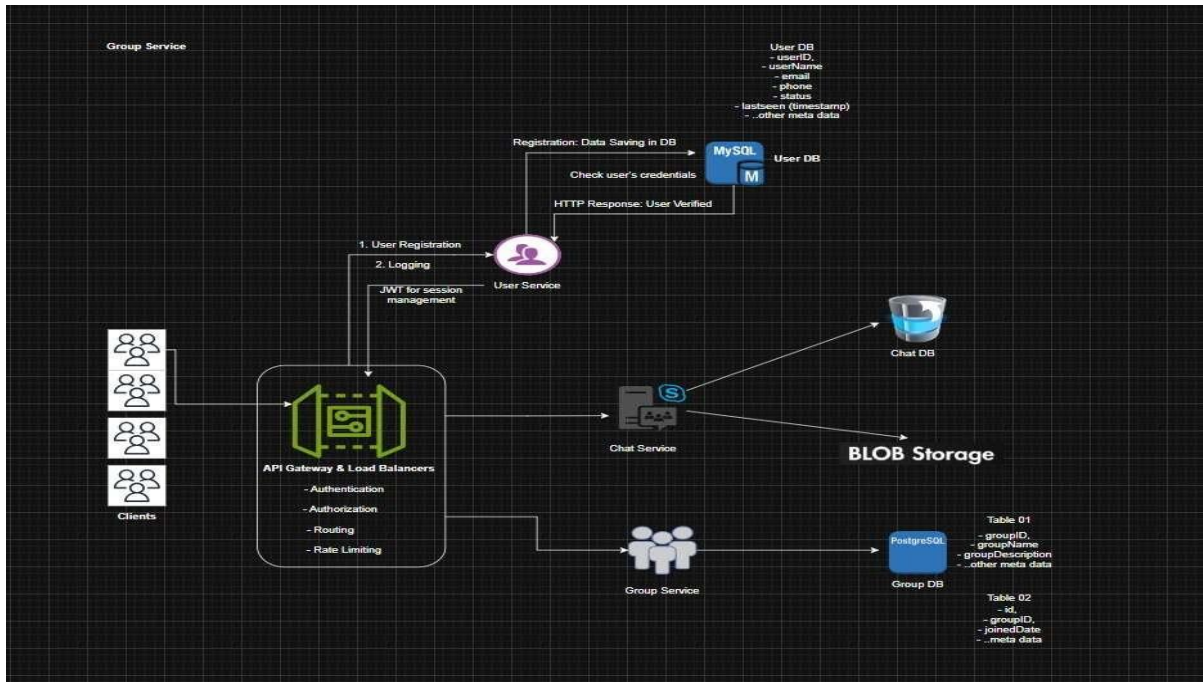
- Scalability
 - o Target: 1 Billion users
 - o 100 messages per user per day
 - o ≈ 100 Billion messages/day
 - o Estimated storage: ~ 100 TB
- CAP Theorem
 - System prioritizes:
 - o High Availability
 - o Partition Tolerance
 - o Eventual Consistency
- Latency: 200-300ms
 - o Target message delivery time: 200–300 ms

5. High Level Design (HLD)

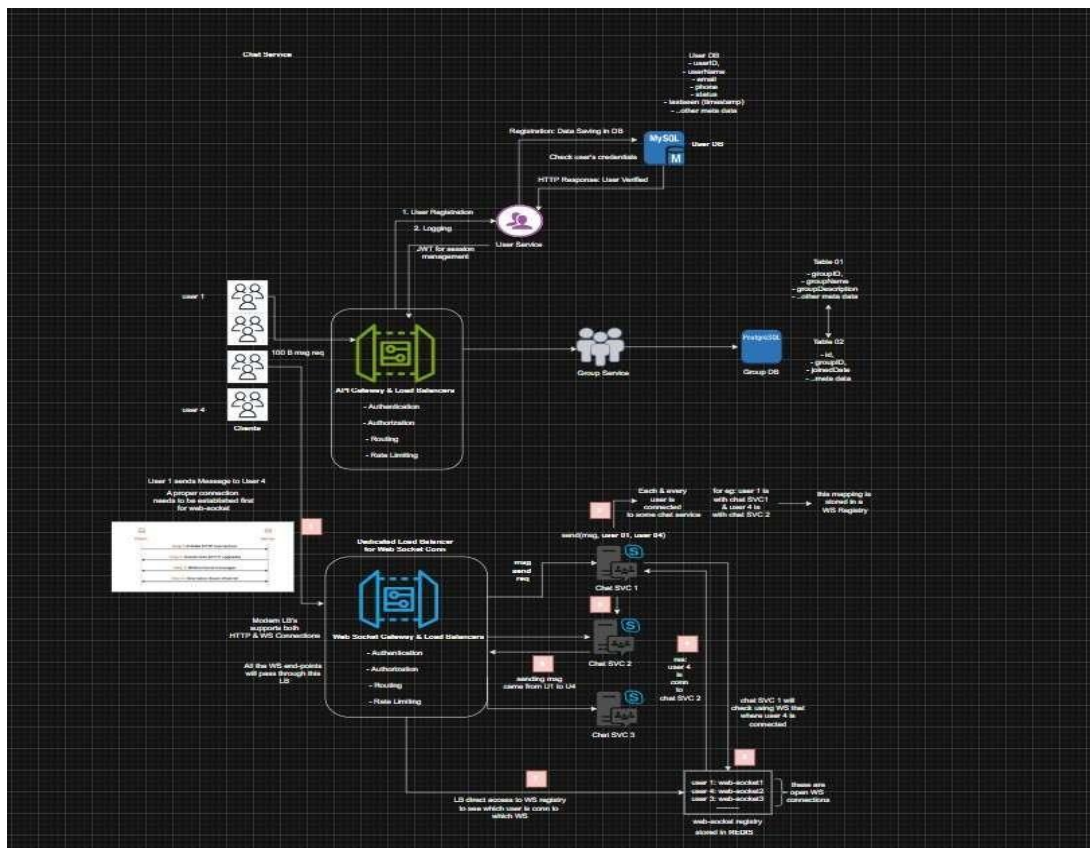
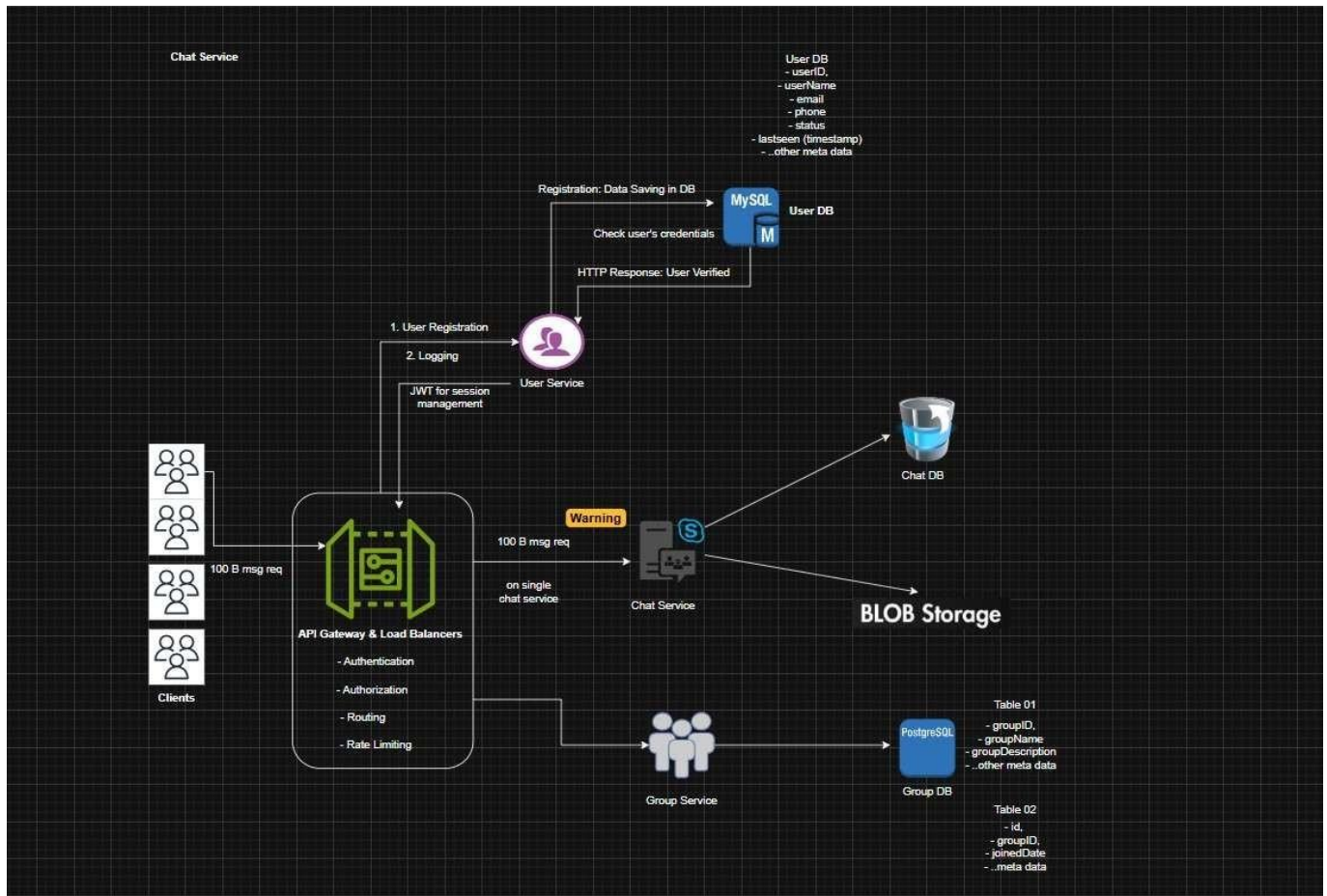


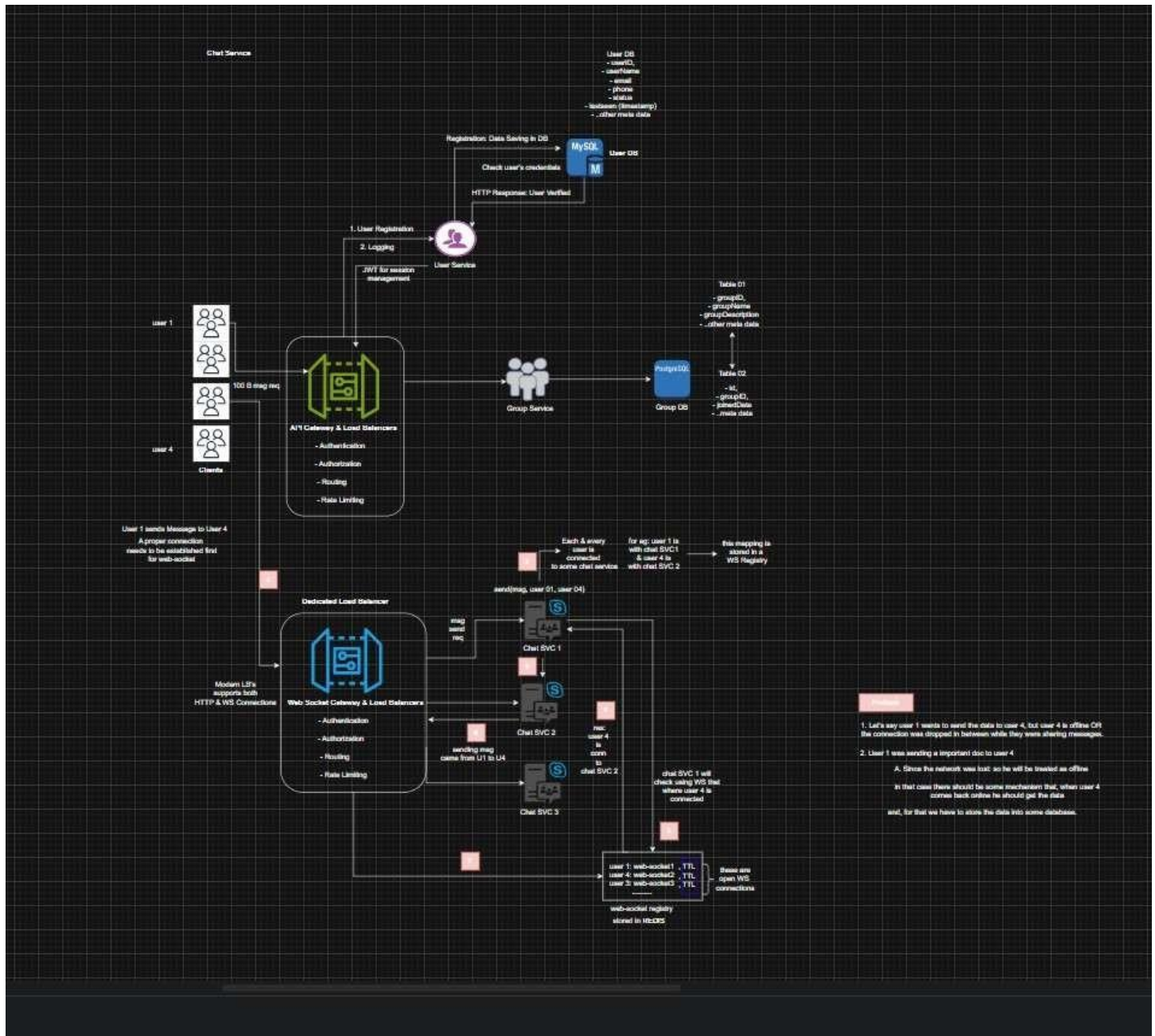
6. Low Level Design (LLD):

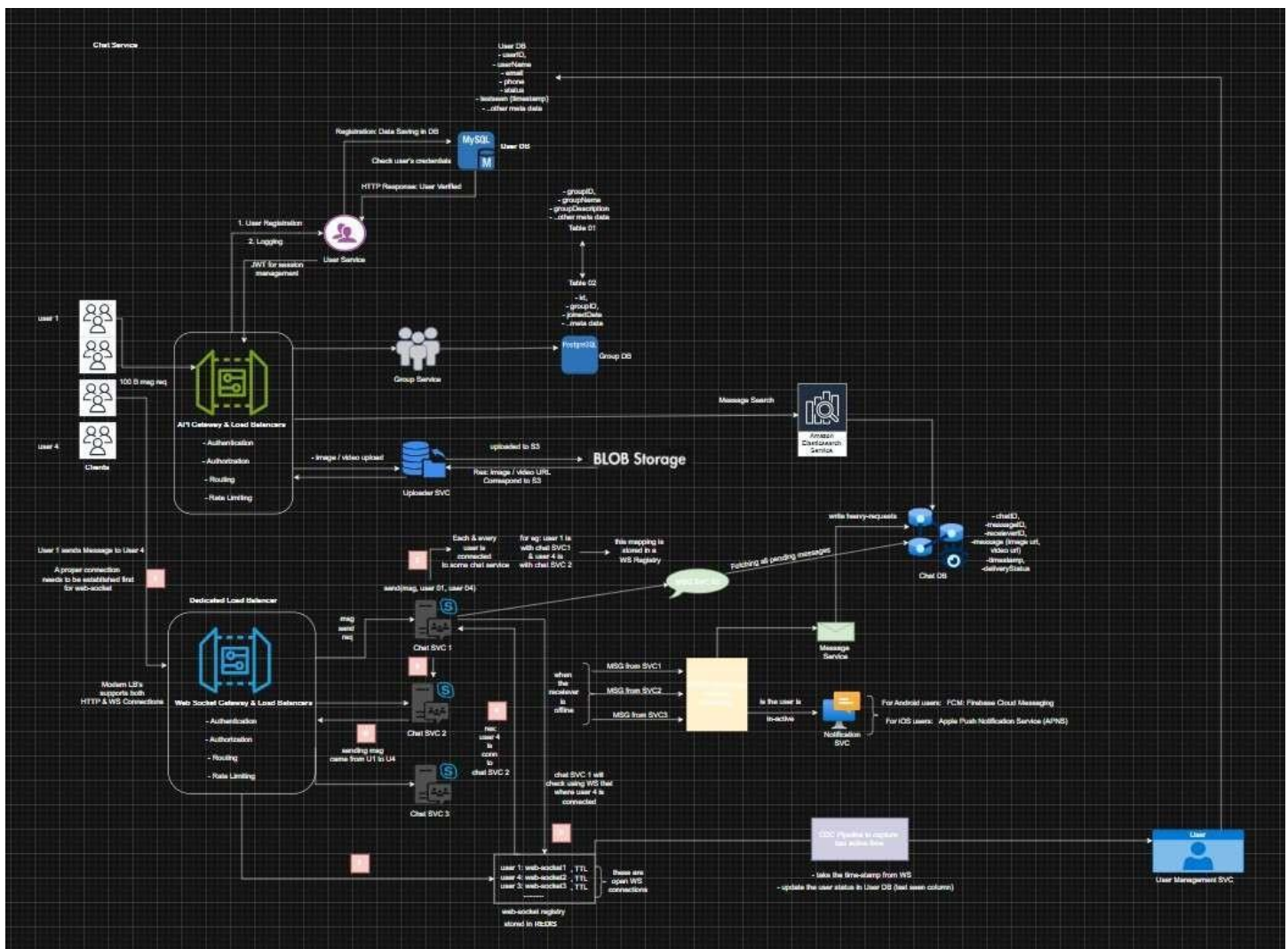
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING







7. Scalability Solution

- **Horizontal Scaling**
Add multiple application servers behind a load balancer to distribute traffic and handle billions of requests efficiently.
- **Database Sharding**
Partition data based on userID or chatID so that message load is distributed across multiple database servers.
- **Caching with Redis**
Store frequently accessed chats and recent messages in Redis to reduce database load and improve response time.
- **Message Queue System**
Use Kafka/RabbitMQ for asynchronous message processing and reliable message delivery.
- **CDN for Media Files**
Use Content Delivery Network to efficiently deliver images, videos, and files globally with low latency.



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

8. Learning Outcomes (What I Have Learnt)

- Understand real-time communication using WebSockets.
- Learn how to design scalable distributed systems.
- Apply CAP Theorem concepts in real-world system design.
- Design RESTful APIs and WebSocket endpoints.
- Analyze trade-offs between consistency, availability, and latency.